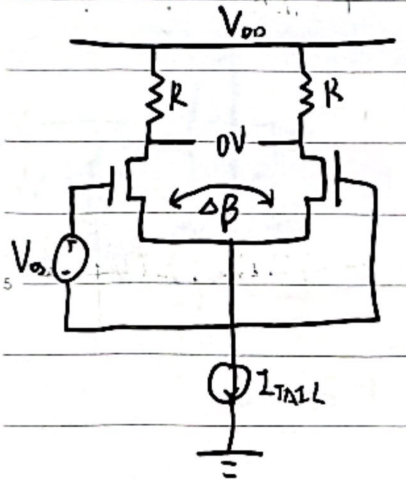
$$\frac{\Delta I_d}{I_D} \approx \frac{g_m V_{os}}{I_D} \approx \frac{\Delta R}{R} \qquad V_{os} = \left(\frac{g_m}{I_D}\right)^{-1} \frac{\Delta R}{R} = \frac{V_{ov}}{2} \cdot \frac{\Delta R}{R}$$

电流不匹配



$$\beta = \mu C_{ox}\frac{W}{L}$$

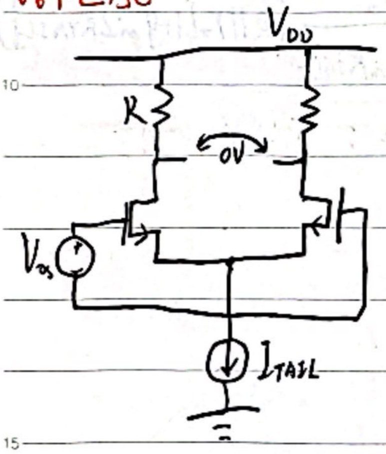
$$\beta V_{ov_1}^2 = (\beta + \Delta\beta) V_{ov_2}^2$$

$$\therefore \frac{V_{ov_1}^2}{V_{ov_2}^2} = 1 + \frac{\Delta\beta}{\beta}$$

$$\frac{V_{ov_1}}{V_{ov_2}} \approx 1 + \frac{V_{os}}{V_{ov}} \approx 1 + \frac{1}{2}\frac{\Delta\beta}{\beta}$$

$$V_{os} \approx \left(\frac{g_m}{I_D}\right)^{-1} \frac{\Delta\beta}{\beta}$$

V_t不匹配



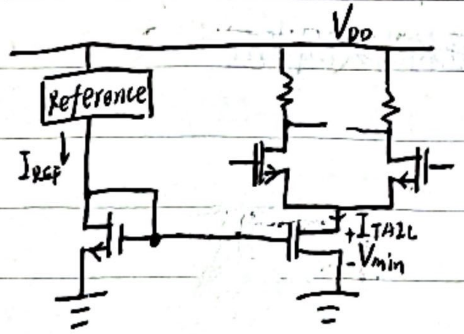
$$V_{os} = V_{t1} - V_{t2} = \Delta V_t$$

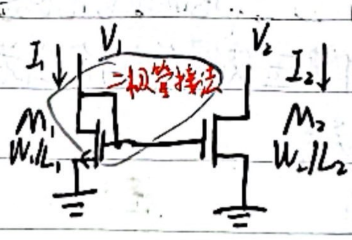
$$\therefore V_{os} \cong \Delta V_t + \left[\frac{g_m}{I_D}\right]^{-1}\left[\frac{\Delta\beta}{\beta} + \frac{\Delta R}{R}\right]$$

$$\sigma_{V_{os}} \cong \sqrt{\sigma_{\Delta V_t}^2 + \left[\frac{g_m}{I_D}\right]^{-2}\left[\sigma_{\frac{\Delta\beta}{\beta}}^2 + \sigma_{\frac{\Delta R}{R}}^2\right]} \qquad \sigma_{\Delta V_t} = \frac{A_{V_t}}{\sqrt{WL}} \qquad \sigma_{\frac{\Delta\beta}{\beta}} = \frac{A_\beta}{\sqrt{WL}}$$

V_t 不匹配是 V_{os} 的主导因素，尤其是在中反型区和弱反型区（$\frac{g_m}{I_D}$较大）

2. 基本电流镜电路





令$L_1 = L_2$

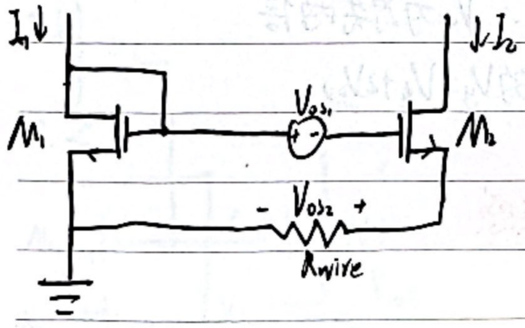
W_1与W_2取整数

$V_{gs} - V_t$相同

$$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}V_{ov}^2$$

$\therefore I$ 取决于 $\frac{W}{L}$

失调电压影响



由于V_t的失配以及接地时可能产生的一些电阻导致失配电压分别为V_{os1}和V_{os2}

$\Delta I = I_1 - I_2 \cong g_m(V_{os1}+V_{os2})$

∴误差为 $\frac{\Delta I}{I} \cong \frac{g_m}{I_o}(V_{os1}+V_{os2})$

∴要想降低误差，需使$\frac{g_m}{I_o}$足够小，则V_{ov}增大，前述V_{min}增大

因为工作在饱和区

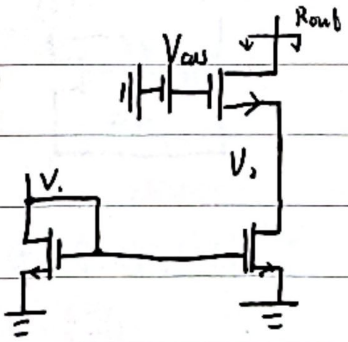
更准确: $$\frac{\Delta I_o}{I_o} = \sqrt{\left(\frac{\Delta \frac{W}{L}}{\frac{W}{L}}\right)^2 + \left(2\frac{\Delta V_t}{V_{gs}-V_t}\right)^2}$$

V_{ds}对电流匹配的影响

由之前的沟道调制效应可知，若V_{ds}存在ΔV_{ds}的差异，则会造成存在$\Delta I = \frac{\Delta V}{r_o}$

为减小ΔI，则需使用r_o较大（$r_o \propto L$，即L较大）的器件，或使两端MOS管的V_{ds}尽可能接近

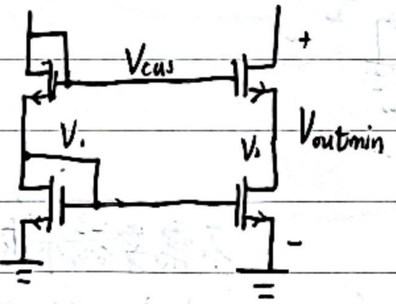
共源共栅电流镜



考虑增加共源共栅级使ΔV_{ds}变小

可知 $R_{out} \cong g_m r_o^2$

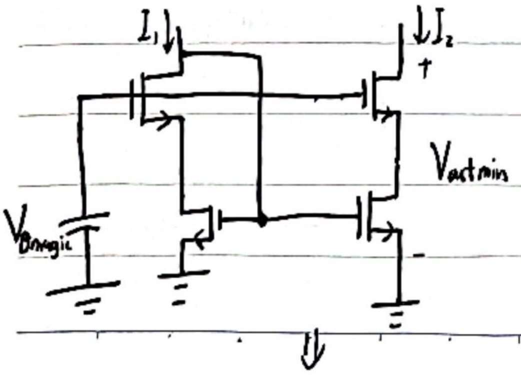
⇒



由图可知，$V_1=V_{gs}=V_2$ ∴ $V_{out\,min}=V_2+V_{DS}=V_2+V_{gs}-V_t=2V_{gs}-V_t=V_t+2V_{ov}$

此时，为使各管都工作在饱和区，会使输出最小电压增大，降低电压裕度

因此为减小输出电压



由图可知 $V_{out\,min}=2V_{ov}$

∴ $V_{Bmagic}=V_{ov}+V_{gs}=V_t+2V_{ov}$

⇓

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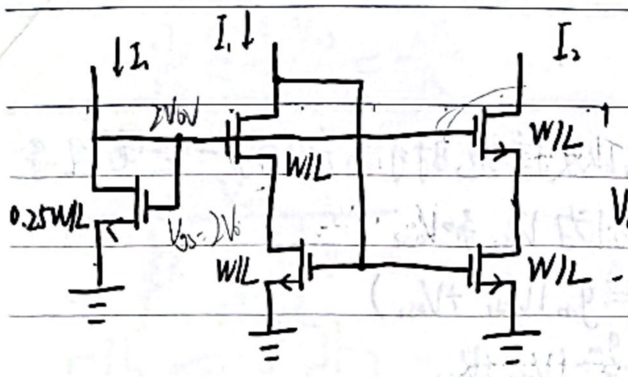
$2V_{ov}$

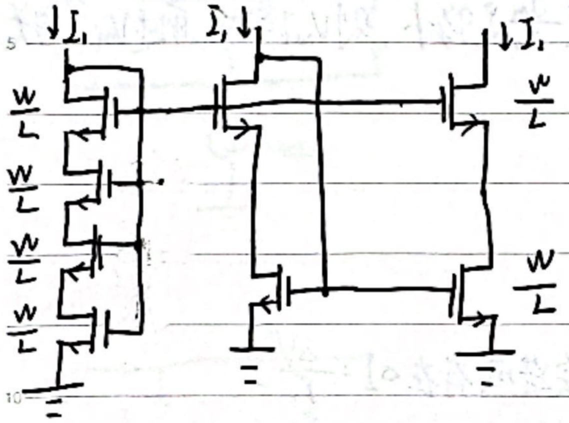


为产生V_{Bmagic}，新增加一个宽长比为原先$\frac{1}{4}$的MOS管

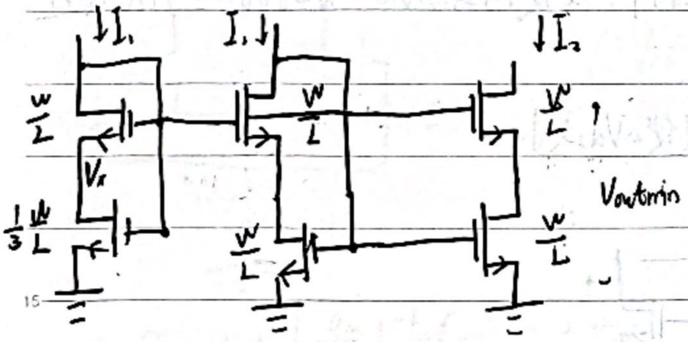
$\because I_1=I_1 \quad \therefore V_{ov}$为原先两倍

因此新管的$V_g=V_t+2V_{ov}$

对体效应敏感



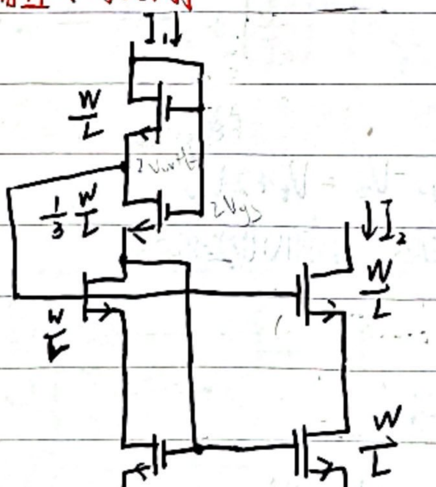
对体效应不敏感，等效为4L



$I_1=\frac{1}{2}\mu C_{ox}\frac{W}{L}V_{ov}^2=\mu C_{ox}\frac{1}{3}\frac{W}{L}\left[(V_x+V_{ov}+V_t-V_0-\frac{V_x}{2})V_x\right]$

$V_x=V_{ov}$

自偏置共源共栅

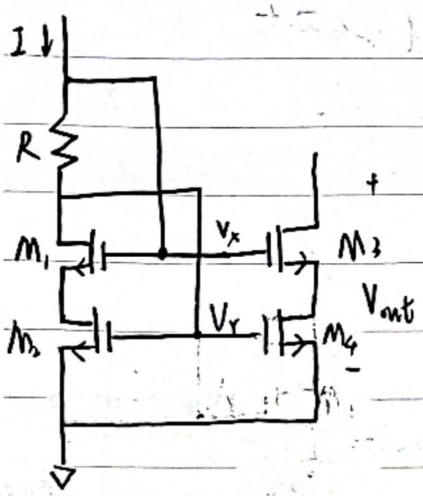


不需要额外电流支路产生V_{cas}偏置电压

对体效应敏感

要求较大的偏置电压裕度

电阻自偏置电流镜



对M_1管，要求处于饱和区

$V_{gs1}-V_{ds1}<V_t$ ∴ $V_X-V_Y<V_t$

∴ $IR<V_t$

对于M_2，要使其工作在饱和区

则 $V_Y-(V_X-V_{gs})<V_t$

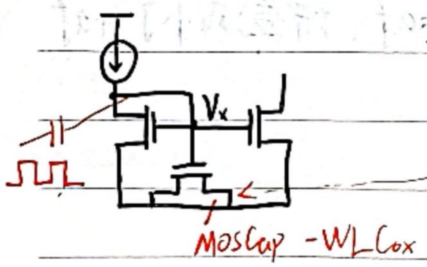
∴ $IR>V_{gs}-V_t$

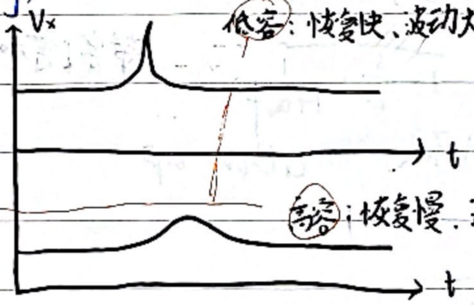
∴ $\frac{V_{gs}-V_t}{I}<R<\frac{V_t}{I}$

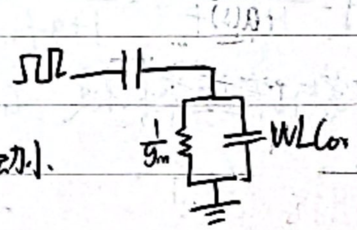
对于X点，有 $V_X=V_{ov}+V_{gs}$ ∴ $V_{out}>V_X+V_t=2V_{ov}$

$R_{out}\approx g_m r_o^2$

3. 电容耦合(capacitive coupling)





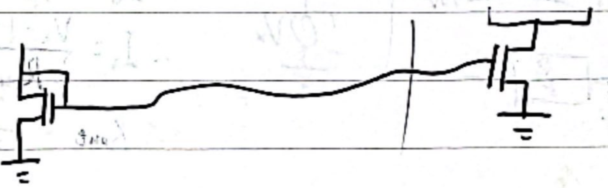


4. 电流镜的两种工作形式

电压传输：

① 有较大的IR压降

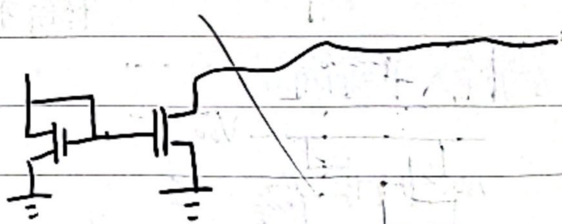
② 通常用于物理上较接近的电路



电流传输：

① 有一个全局参考电流

② 消耗额外电流



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工艺偏差与反馈

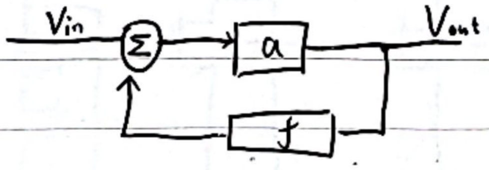
1. PVT:

Process：加工过程中产生的偏差

Voltage：电压带有±10%的偏差

Temprature：温度引起的偏差

2. 负反馈：



$V_{out}=a(V_{in}-fV_{out})$

$\frac{V_{out}}{V_{in}}=\frac{a}{1+af}$

当 $af>>1$ 时，$\frac{V_{out}}{V_{in}}=\frac{1}{f}$　　当a足够大使 $af>>1$ 时，$\frac{V_{out}}{V_{in}}$ 只取决于 f

定义 $A=\frac{a}{1+af}$　　$\frac{\Delta A}{A}=\frac{\frac{\Delta a}{a}}{1+af}=\frac{\frac{\Delta a}{a}}{1+T}$，$T=af$（环路增益），T越大，敏感度越低

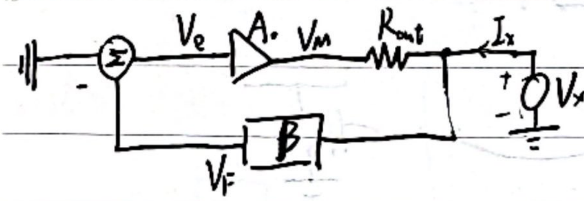
负反馈抑制了高次分量，提高了线性度

带宽　[a 变为 $a(s)$　$a(s)=\frac{a_0}{1-\frac{s}{P_1}}$]

此时 $A=\frac{a(s)}{1+a(s)f}=\frac{a_0}{1+a_0f}\cdot\frac{1}{1-\frac{s}{P_1}\cdot\frac{1}{1+a_0f}}$　　带宽增加了 $1+af$，增益减小了 $1+af$

※即带宽与增益乘积不变，GBW不变　　$GBW=aP$

输出阻抗



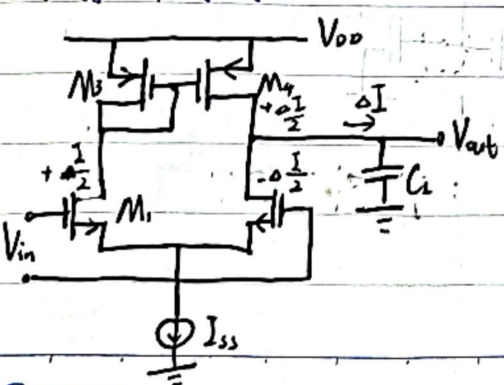
$V_e=-V_F=-\beta V_x$

$\therefore I_x=\frac{V_x-(-\beta V_x)}{R_{out}}=\frac{(1+\beta A_0)V_x}{R_{out}}$

$\therefore R_{out}'=\frac{V_x}{I_x}=\frac{R_{out}}{1+\beta A_0}$

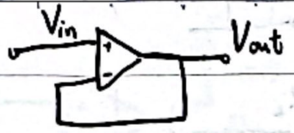
单端和全差分电路

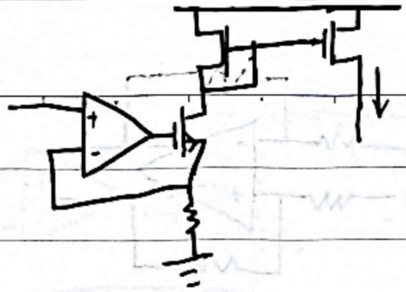
1. 差分输入单端输出



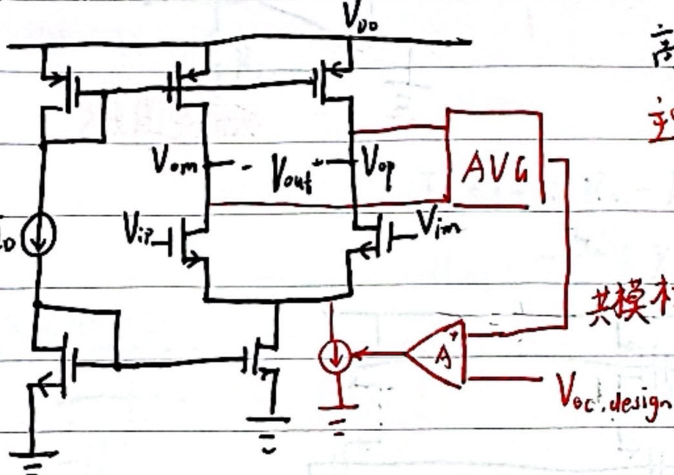
$A=g_m(r_{on}\parallel r_{op})$

2. 单位增益缓冲器





3. 差分输入差分输出

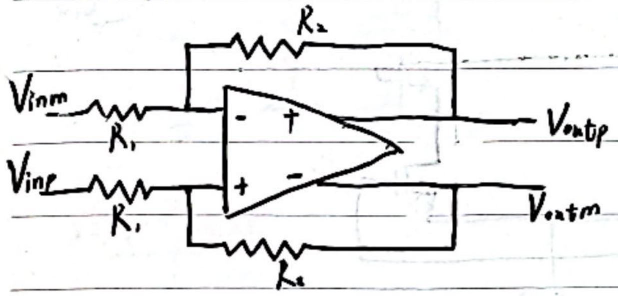


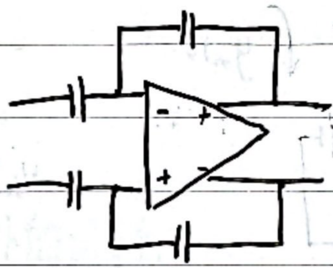
高阻点，共模电压很难确定

主要是由于上下电流镜失配导致出现输出共模偏移

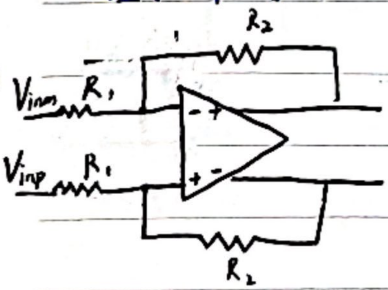
$A = g_m(r_{on} \| r_{op})$

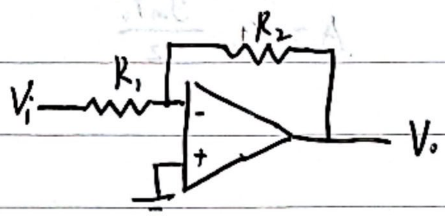
4. 全差分电路





5. 全差分与单端输入运放的差别





对称	低复杂度
可以抑制噪声	不需要共模反馈
good PSRR	少反馈组件
容易分析	无需其他器件可构成缓冲器
输出幅度是单端的两倍	

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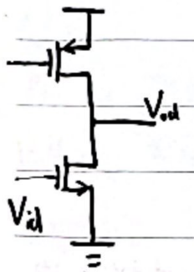
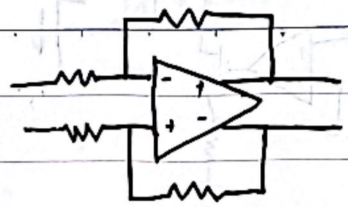
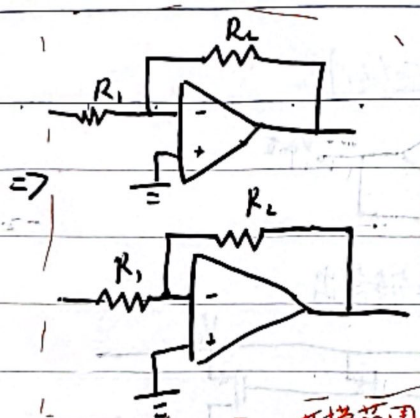
6、全差分电路等效电路



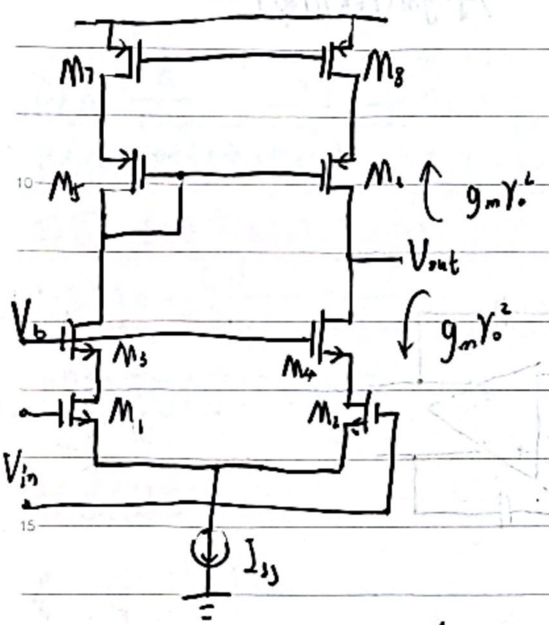


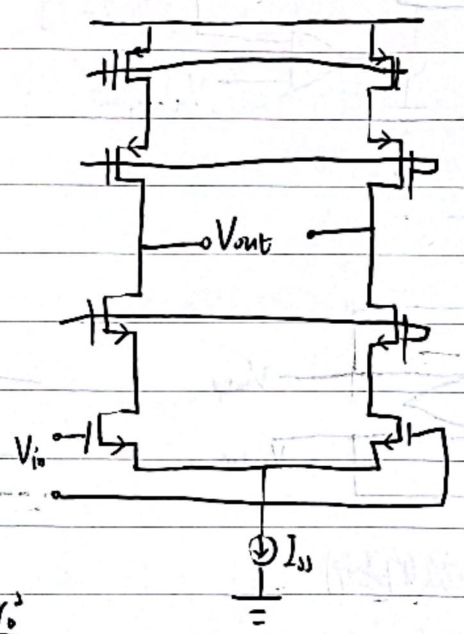


共模范围更大

7.运算放大器拓扑结构

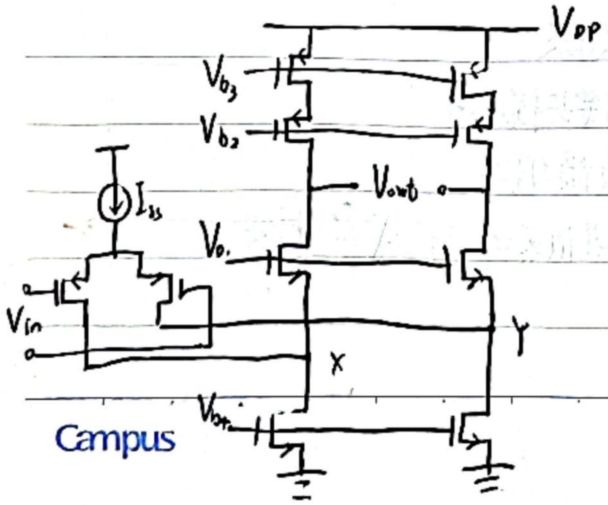
套筒式共源共栅放大器





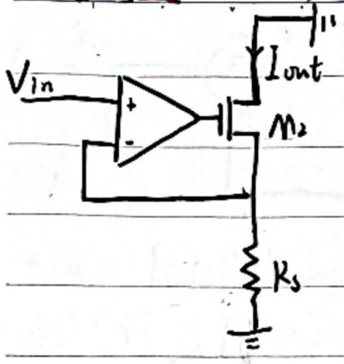
$$A = g_m \frac{g_m r_o^2}{2}$$

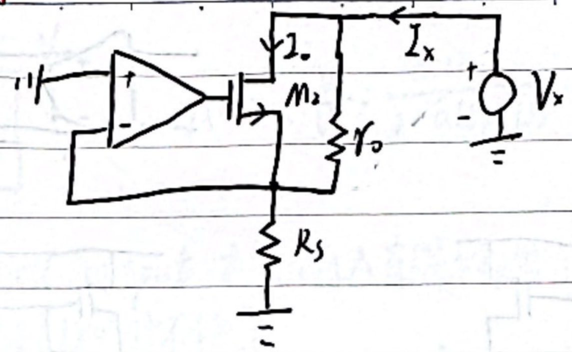
折叠式共源共栅



$$A = g_m\left[(g_m r_o^2) \| \left(\frac{g_m r_o^2}{2}\right)\right]$$

自增增益　进一步增大输出阻抗

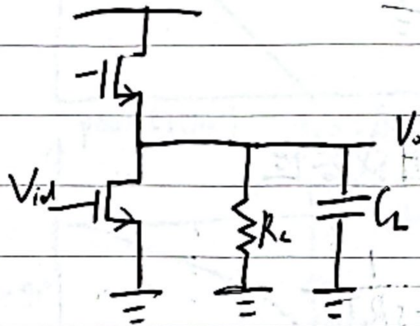




$$I_x = (-A_1 R_s - R_s) g_m I_x + \frac{V_x - R_s I_x}{r_o}$$

$$R_{out} = \frac{V_x}{I_x} = r_o + (A_1 + 1) g_m r_o R_s + R_s$$

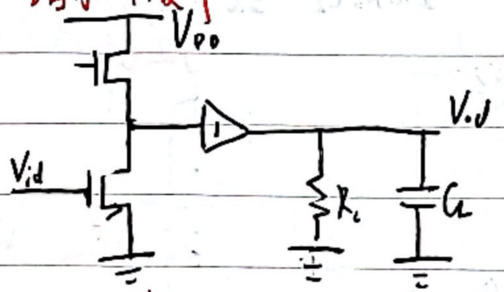
8. 负载的选择



R_L过小会导致增益损失

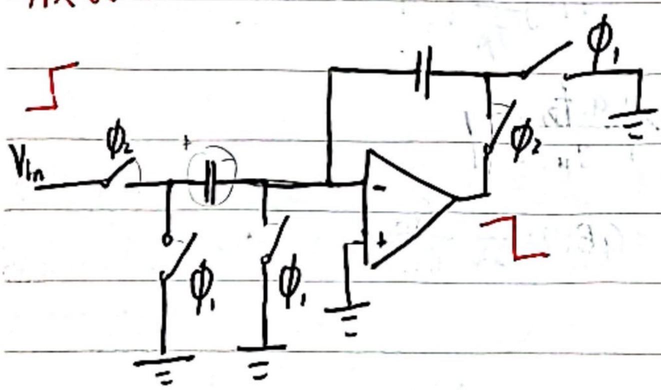
$\Downarrow$

增加一个缓冲



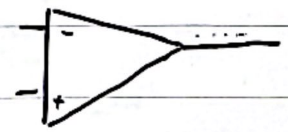
需额外的功耗

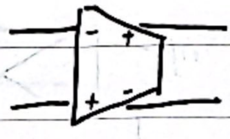
开关电容电路

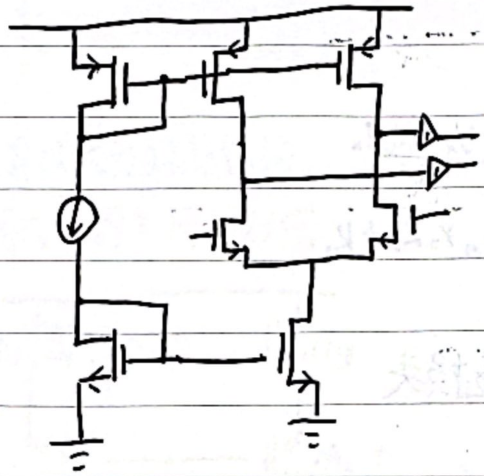


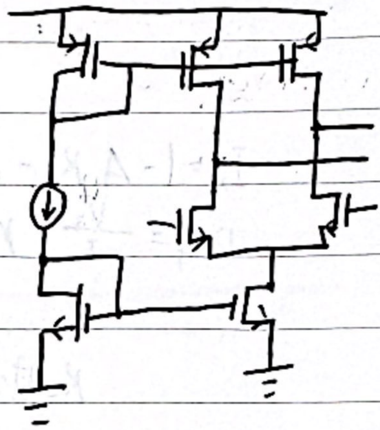
ϕ_1用于初始化电容

9. 运放与跨导放大器





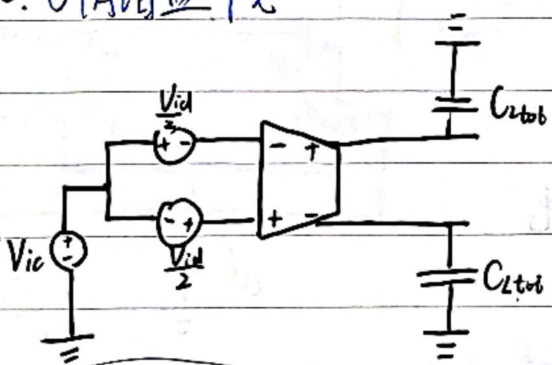




比较通用

只能驱动高阻或容性

10. OTA增益带宽



$$\frac{V_{od}}{V_{id}} = -G_m R_o \frac{1}{1+sR_oC_{Ltot}}$$

$$GBW = G_m R_o \cdot \frac{1}{2\pi R_o C_L} = \frac{1}{2\pi}\frac{G_m}{C_{Ltot}}$$

与R_o无关

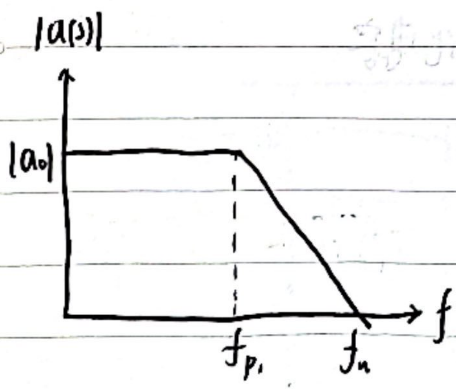
$$\frac{R_o \cdot \frac{1}{sC_{Ltot}}}{sC_{Ltot}R_o + \frac{1}{sC_{Ltot}}}$$

单位增益频率

$$a(s) = \frac{V_{od}}{V_{id}} = -G_m R_o \cdot \frac{1}{1+sR_oC_{Ltot}} = \frac{a_o}{1-\frac{s}{p_1}}$$

令 $\left|\frac{a_o}{1-j\frac{f_u}{f_{p_1}}}\right| = 1$

$\left|\frac{a_o f_{p_1}}{f_u}\right| = 1$



$$f_u \gg f_{p_1} \Rightarrow \left|\frac{a_o f_{p_1}}{f_u}\right| = 1$$

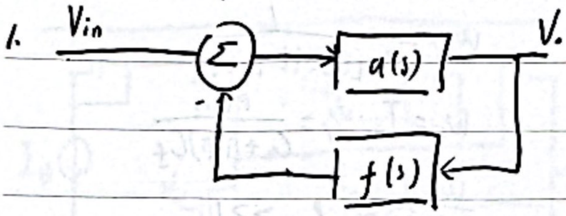
$|a_o| f_{p_1}$

$$f_u = |a_o| f_{p_1} = GBW = \frac{1}{2\pi}\frac{G_m}{C_{Ltot}}$$

$f_u = |a_o| f_p$

反馈电路稳定性分析

1.

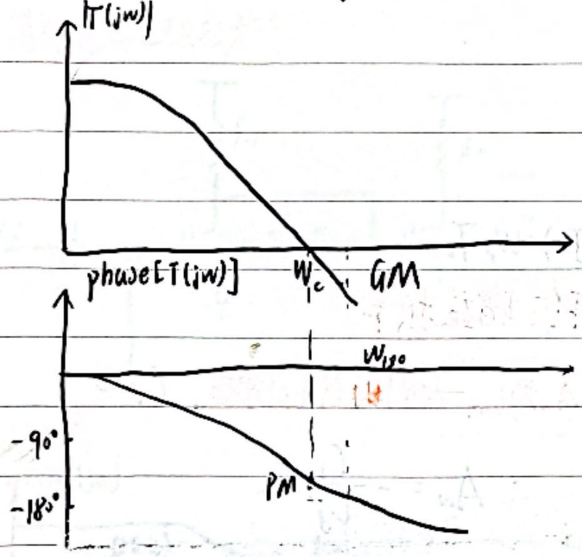


$$A(s)=\frac{V_o}{V_i}=\frac{a(s)}{1+a(s)f(s)}=\frac{a(s)}{1+T(s)}$$

BIBO: bounded input - bounded output 有限输入有限输出

波特准则: 当 $|T(j\omega)|>1$ 且相移达到 180° 时不稳定

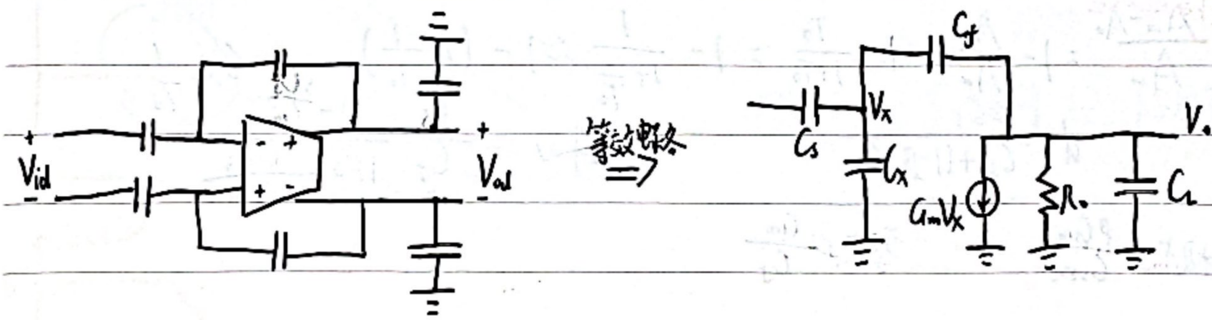
相位裕度: $PM=180^\circ-Phase[T(j\omega)]_{\omega=\omega_c}$ ω_c 为0dB(无增益)对应频率



$$GM=\frac{1}{|T(j\omega)|}\Big|_{\omega=\omega_{180}}$$

通常 $GM\geq 3\ldots5$

2. 电容反馈OTA



反馈比例分析

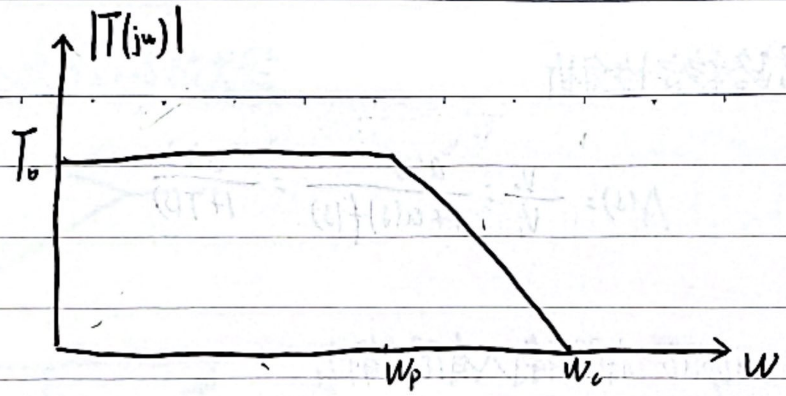
① 所有独立源置0　② 断开反馈环路中的一个受控源

③ 在断点注入一个检测信号 s_t　④ 找出由断开受控源返回的信号 s_r

⑤ 环路增益 $T=-\frac{s_r}{s_t}$

$V_x=\beta V_o$　$\beta=\frac{C_f}{C_s+C_x+C_f}$ 反馈系数　$V_o=-i_t\cdot\left(R_o\|\frac{1}{sC_{tot}}\right)$　$C_{tot}=C_L+(1-\beta)C_f$

$i_r=G_mV_x$　$T(s)=-\frac{i_r}{i_t}=\beta\cdot G_m\cdot\left[R_o\|\frac{1}{sC_{tot}}\right]=\beta\cdot G_m\cdot R_o\cdot\frac{1}{1+sR_oC_{tot}}$



$T_0 = \beta G_m R_o$

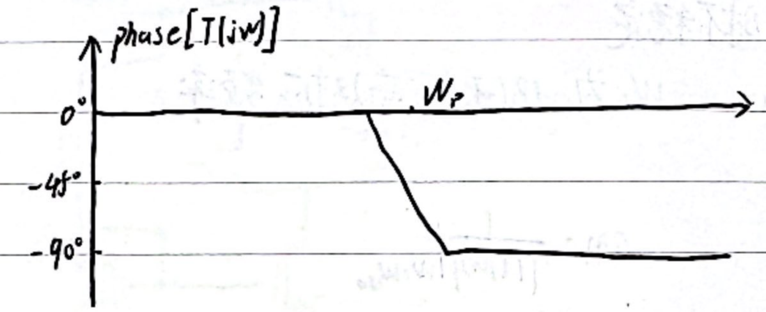
$\omega_p = \frac{1}{R_o[C_L+(1-\beta)C_f]}$

$\omega_c = T_0 \cdot \omega_p = \frac{\beta G_m}{C_L+(1-\beta)C_f}$

$\frac{\omega_c}{\omega_p} \approx \beta G_m R_o >> 1$

$phase[T(j\omega)]|_{\omega=\omega_c} = -\tan^{-1}\left[\frac{\omega_c}{\omega_p}\right] = -90^\circ$

$PM = 90^\circ$



反馈使带宽增加到(1+af)　　$\therefore \omega_{-3dB} = \omega_p(1+T_0) \approx \omega_p T_0$

$\therefore \omega_{3dB} = \frac{\beta G_m R_o}{R_o[C_L+(1-\beta)C_f]} \approx \omega_c$　　闭环带宽等于单位增益频率

$A(s) = \frac{V_o}{V_i} = A_\infty \frac{T(s)}{1+T(s)} + \frac{d}{1+T(s)}$　　构造出来的　　A_∞为$G_m \to \infty$时的环路增益，$G_m \to 0$

基于环路增益的闭环增益

当$G_m \to \infty$时，$V_x = 0$　　$\therefore 0 = V_i sC_s + V_o sC_f$　　$\therefore A_\infty = -\frac{C_s}{C_f}$

低频时，$d_0 = 0 \Rightarrow A_0 = -\frac{C_s}{C_f}\frac{T_0}{1+T_0}$

定义 $\varepsilon = \frac{A_\infty - A_0}{A_\infty} = 1 - \frac{A_0}{A_\infty} = 1 - \frac{T_0}{1+T_0} = 1 - \frac{1}{1+\frac{1}{T_0}} \approx 1 - (1 - \frac{1}{T_0})$　　$\varepsilon = \frac{1}{T_0}$

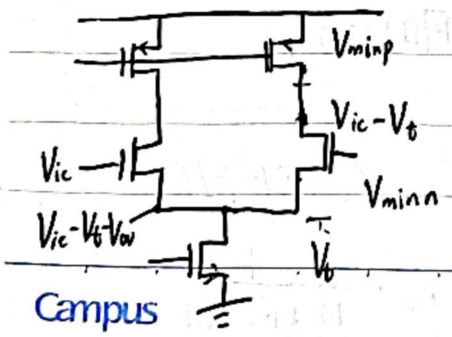
静态增益误差

高频时　$d = \frac{\beta C_s}{C_L+(1-\beta)C_f}$　　$A(s) = -\frac{C_s}{C_f}\frac{1-\frac{sC_f}{G_m}}{1+s\frac{C_{Ltot}}{\beta G_m}}$

极点 $s = \frac{\beta G_m}{C_{Ltot}}$　　零点 $s = \frac{G_m}{C_f}$

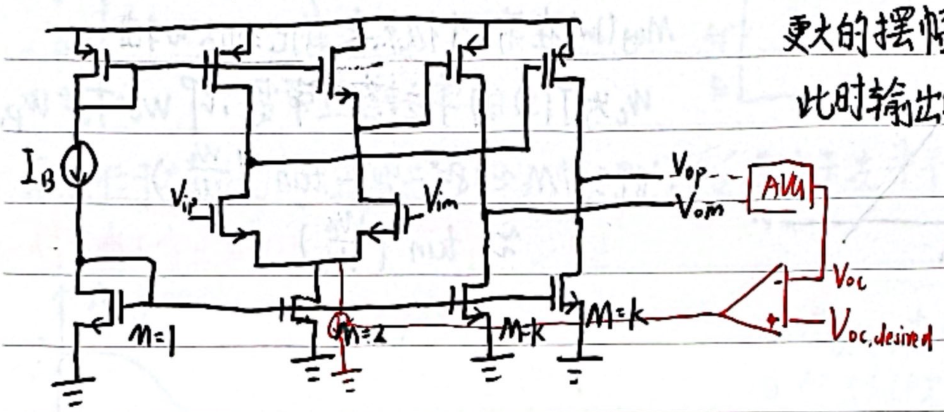
二级放大器

1. OTA输出摆幅分析



$V_{odpp\,max} = 2(V_{DD} - V_{minp} - V_{minn} - V_{mint})$

2. 二级放大器

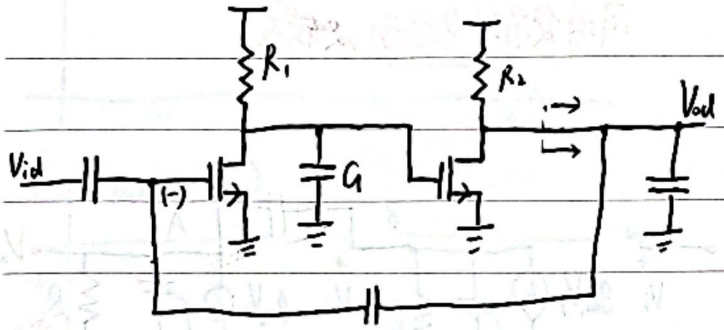


更大的摆幅 更高的增益 $(\frac{g_m r_o}{2})^2$

此时输出的摆幅不再受限于输入

通过共模反馈，使 $V_{oc}=V_{oc,desired}$

简化闭环二级放大器

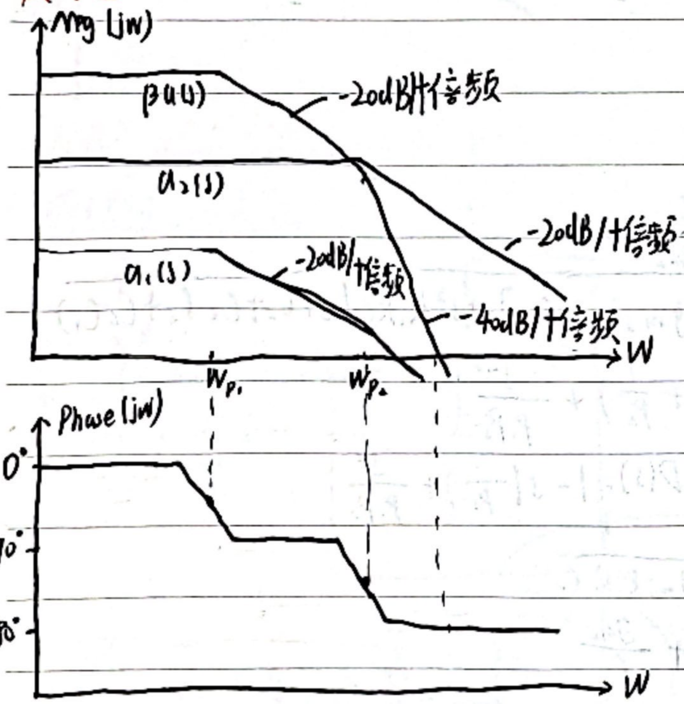


$$T(s)=\beta\frac{g_{m1}R_1 g_{m2}R_2}{(1-\frac{s}{P_1})(1-\frac{s}{P_2})}=\beta a_1(s)a_2(s)=\beta a(s)$$

$P_1=-\frac{1}{R_1C_1}$ $\quad$ $P_2=-\frac{1}{R_2C_2}$

双极点

波特图：



如果 W_{p1} 与 W_{p2} 很接近，则当 Mg(jw) =0dB时

Phase(jw)=-180°，PM=0，不稳定

∴当2个极点靠太近时，相位裕度不足

引入主极点

为了使电路稳定，需有一个为主极点，且两极点距离足够大

No.

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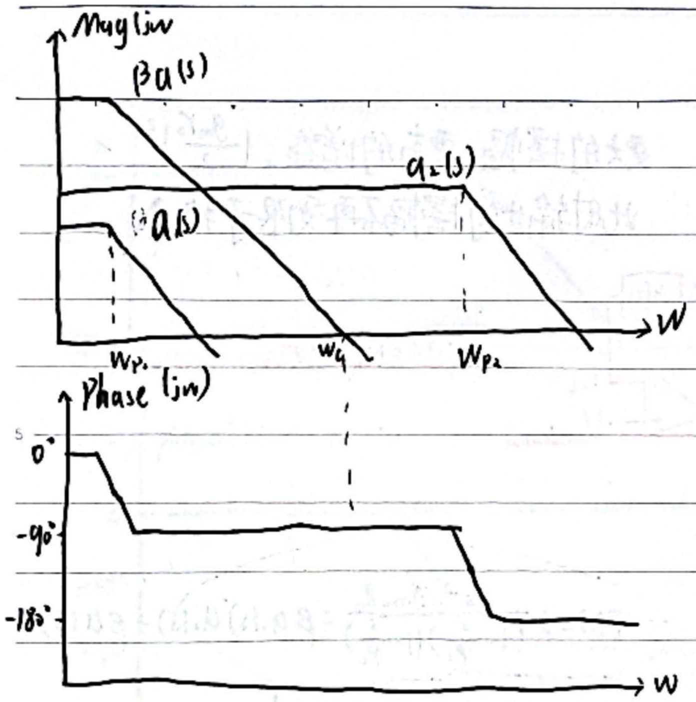


Mag(jω)在第二个极点之前已到达ω轴

ω_c为$T(s)$的单位增益带宽，即 $\omega_c = T_0 \cdot \omega_{p_1}$

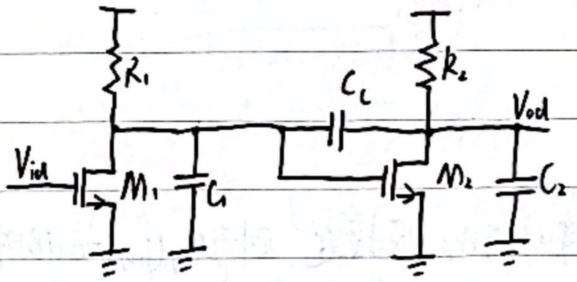
此时 $PM \approx 180° - 90° - \tan^{-1}\left(\frac{\omega_c}{\omega_{p_2}}\right)$

$\approx \tan^{-1}\left(\frac{\omega_{p_2}}{\omega_c}\right)$

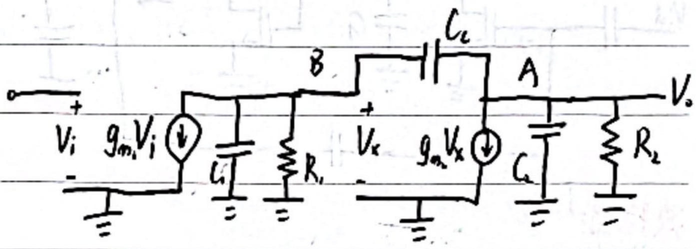
此时的问题为ω_c越小，闭环带宽越小，

同时使用的电容面积较大

密勒补偿



小信号 ⇒



对A、B两点分别列KCL方程可解得

$$a(s) = \frac{V_o}{V_i} = \frac{g_{m1}R_1 g_{m2}R_2\left(1 - s\frac{C_c}{g_{m2}}\right)}{1 + s\left[(C_2+C_c)R_2 + (C_1+C_c)R_1 + g_{m2}R_1R_2C_c\right] + s^2R_1R_2(C_1C_2 + C_cC_2 + C_cC_1)}$$

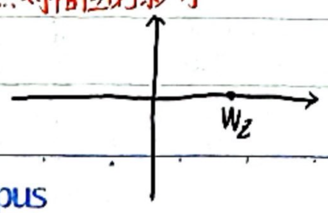
对分母，记 $D(s) = \left(1 - \frac{s}{p_1}\right)\left(1 - \frac{s}{p_2}\right) = 1 - s\left(\frac{1}{p_1} + \frac{1}{p_2}\right) + \frac{s^2}{p_1p_2}$

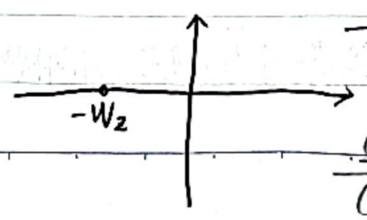
根据最终效果 $p_2 >> p_1$ $\therefore \frac{1}{p_1} >> \frac{1}{p_2}$ $\therefore D(s) = 1 - s\left(\frac{1}{p_1}\right) + \frac{s^2}{p_1p_2}$

则 $p_1 \approx \frac{1}{(C_2+C_c)R_2 + (C_1+C_c)R_1 + g_{m2}R_1R_2C_c} \approx \frac{1}{g_{m2}R_1R_2C_c}$

$\therefore p_2 \approx \frac{g_{m2}}{\frac{C_1C_2}{C_c} + C_1 + C_2}$ 同时零点 $z = +\frac{g_{m2}}{C_c}$ $\frac{1}{-g_{m2}R_1R_2C_c}$

正负零点对相位的影响

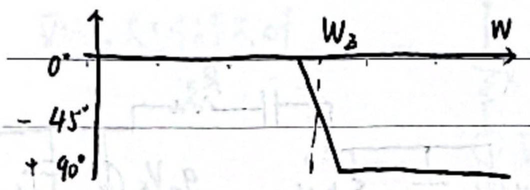


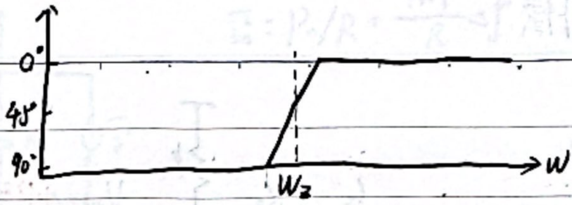


$\frac{g_{m2}}{\frac{C_1C_2}{C_c} + C_1 + C_2}$

$\frac{g_{m2}}{C_c}$

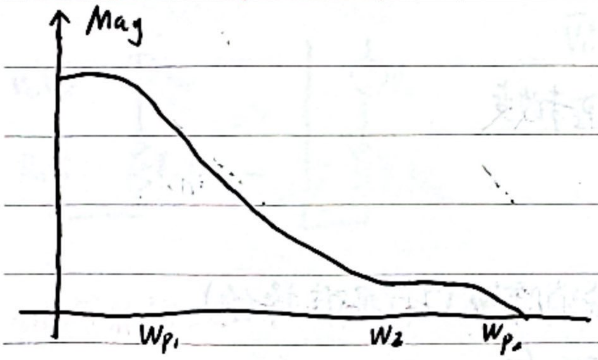
Campus





零点位于右半平面时(正零点),相位滞后90°,零点位于左半平面时(负零点),相位超前90°

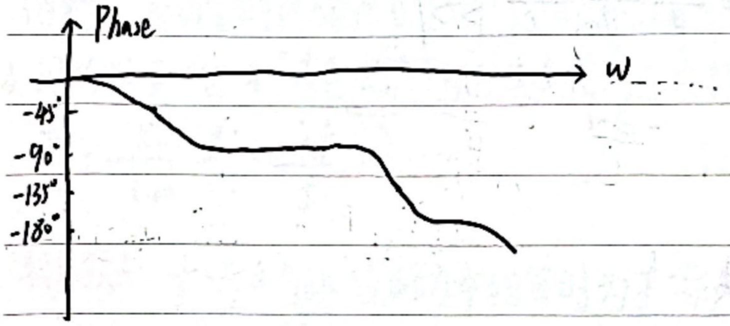
对含有正零点的OTA频率分析



当W_z在W_c左侧时,由于z使相位有90°的下降,但幅频却不下降,大大恶化了PM,
所以只有$W_z \gg W_c$时才可忽略W_z的影响

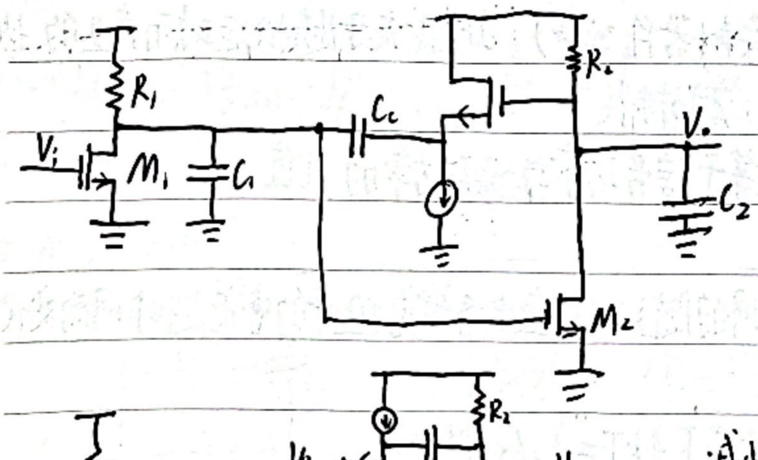
$$W_c = T_0 \cdot W_{p1} \approx \beta \frac{g_m}{C_c}$$

$$\frac{W_z}{W_c} = \frac{1}{\beta} \cdot \frac{g_{m2}}{g_{m1}}$$



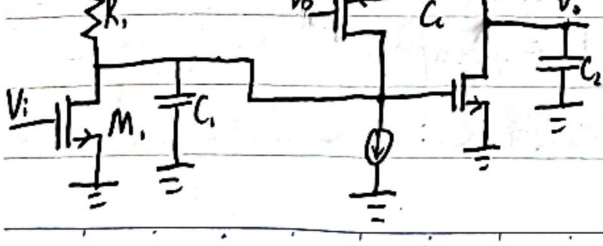
消除正零点的影响

源级跟随器补偿



前馈减小,零点消除

减小了摆幅, $V_{min} = V_{gs}+V_{ov}$, 原先为V_{ov}

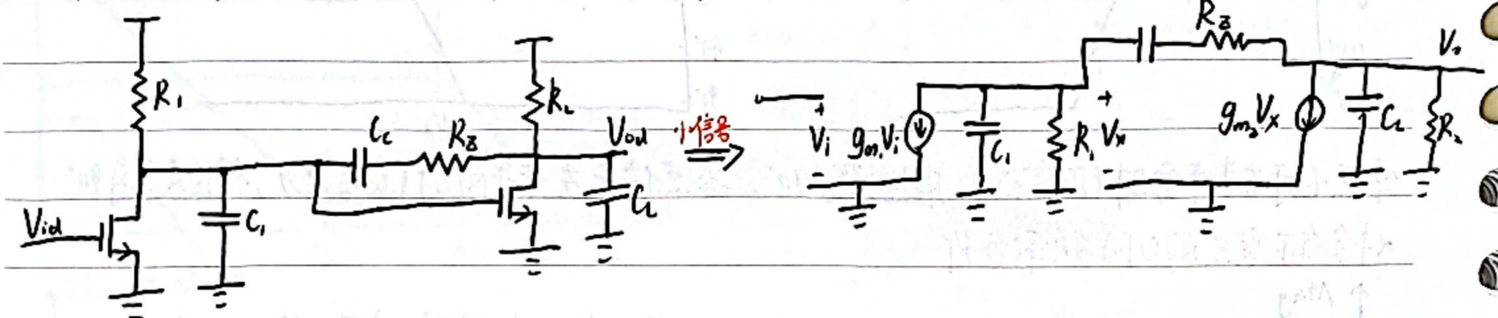


减小前馈

No.

Date

调零电阻补偿



此时 $a(s)\approx a_{v_0}\cdot\dfrac{1-sC_c(\frac{1}{g_{m_2}}-R_z)}{(1-\frac{s}{p_1})(1-\frac{s}{p_2})(1-\frac{s}{p_3})}$ 其中 P_1、P_2 未改变

$-sC_c(\frac{1}{g_{m_2}}-R_z)$

此时，若 $R_z=\dfrac{1}{g_{m_2}}$ 时，W_z 被推向 ∞

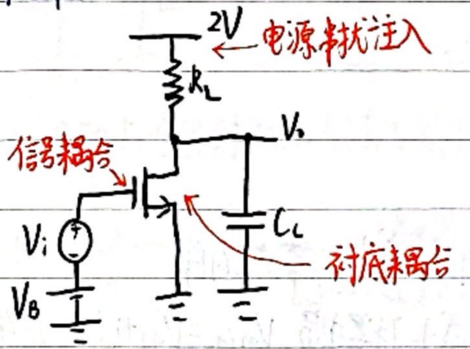
若 $R_z=\dfrac{1+\frac{C_2}{C_c}}{g_m}$ 时，若 $C_2>C_1$，则 W_z 近似 W_{p_2} 此时可能消除 W_{p_2}（实际很难操作）

对于 W_{p_3}，$W_{p_3}\approx\dfrac{1}{R_zC_1}\approx\dfrac{g_{m_2}}{C_1}$ $\qquad \dfrac{W_{p_3}}{W_c}=\dfrac{1}{\beta}\dfrac{g_{m_2}}{g_{m_1}}\dfrac{C_c}{C_1}>>1$

$\dfrac{\frac{g_{m_2}}{C_1}}{\frac{g_{m_1}}{C_c}}$ $\qquad \dfrac{g_{m_2}}{g_{m_1}}\dfrac{C_c}{C_1}$

噪声

1. 噪声来源



种类：

人为噪声：大多由于电路（包括layout）的设计所导致。主要产生原因有：信号耦合、衬底耦合、电源串扰等 用全差分电路解决

固有噪声（器件噪声）：由于载流子随机运动而产生的 热噪声；由于材料

2. 信噪比：衡量有用信号中噪声的大小，数值等于信号功率与噪声功率的比值，

$$SNR=\frac{P_{signal}}{P_{noise}}\propto\frac{V^2_{signal}}{P_{noise}}$$

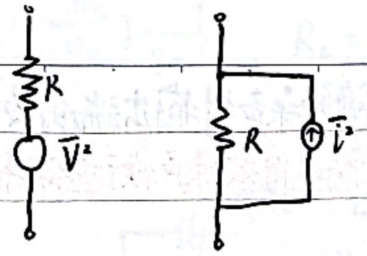
3. 热噪声：在实际电阻中，由于载流子与晶格之间的随机碰撞会导致电阻中的电流随时间而变化，形成时变的电流，称为"热噪声"

功率谱密度 $\quad PSD(f)=n_0=4kT \qquad$ 常温下 $4kT=1.66\times10^{-20}J$

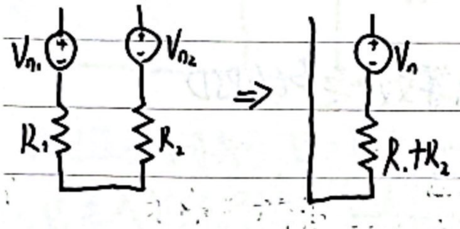
电阻在特定频带内的总噪声平均能量为 $P_n=\int_{f_1}^{f_2}4kT\,df=4kT\Delta f$

等效噪声源：可用等效电压源或等效电流源对噪声源进行等效

$\overline{V^2} = P_n \cdot R = 4kT \cdot R\Delta f$ $\overline{I_n^2} = P_n/R = \frac{4kT}{R}\Delta f$



独立噪声源：



$\overline{V_n^2} = \overline{(V_{n1}-V_{n2})^2} = \overline{V_{n1}^2} + \overline{V_{n2}^2} - 2\overline{V_{n1}\cdot V_{n2}}$

$\overline{V_n^2} = \overline{V_{n1}^2} + \overline{V_{n2}^2} = 4kT(R_1+R_2)\Delta f$ 独立噪声源应用均方值相加

MOS管的热噪声

饱和区，MOS管的热噪声可用一个谱密度为 $\overline{I_n^2} = 4kT\gamma g_m \cdot \Delta f$，对于长沟道，$\gamma = \frac{2}{3}$

4. 闪烁噪声：也叫"有色噪声"与白噪声相对，白噪声指功率谱不随频率变化的噪声，如热噪声

$$\overline{I_f^2} = \frac{K_f}{C_{ox}}\frac{g_m^2}{W\cdot L}\cdot\frac{\Delta f}{f} = PSD(f)$$

转角频率：$\frac{1}{f}$噪声与热噪声功率相等时为$\frac{1}{f}$噪声的"转角频率"

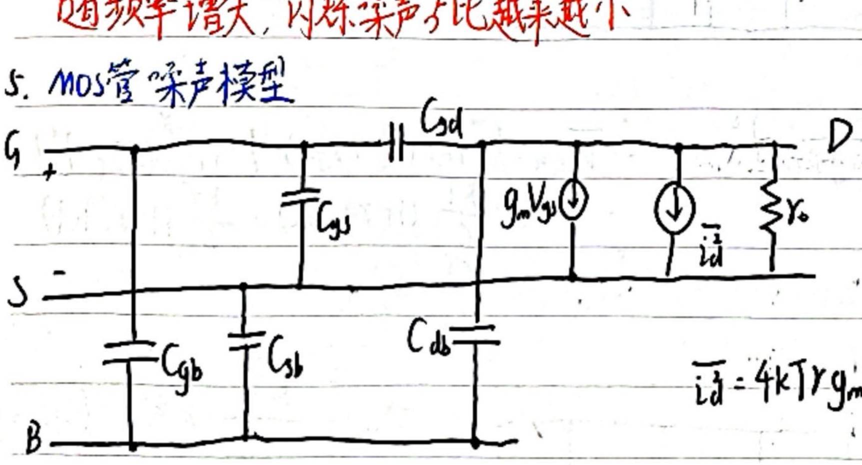
$$\frac{K_f}{C_{ox}}\frac{g_m^2}{WL}\cdot\frac{\Delta f}{f_{co}} = 4kT\cdot\gamma\cdot g_m\cdot\Delta f \Rightarrow f_{co} = \frac{K_f}{4kT\gamma}\frac{1}{C_{ox}}\frac{g_m}{WL}$$

$$= \frac{K_f}{4kT\gamma}\frac{1}{C_{ox}}\frac{1}{L}\cdot\left(\frac{g_m}{I_D}\right)\left(\frac{I_D}{W}\right)$$

总功率分布：$\overline{i_{f,tot}^2} = \int_{f_1}^{f_2}\frac{K_f}{C_{ox}}\cdot\frac{g_m^2}{WL}\cdot\frac{\Delta f}{f} = \frac{K_f}{C_{ox}}\cdot\frac{g_m^2}{WL}\ln\frac{f_2}{f_1} = \frac{K_f}{C_{ox}}\cdot\frac{g_m^2}{WL}2.3\log\frac{f_2}{f_1}$

随频率增大，闪烁噪声占比越来越小

5. MOS管噪声模型



$$\overline{i_d^2} = 4kT\gamma g_m\cdot\Delta f + \frac{K_f}{C_{ox}}\cdot\frac{g_m^2}{W\cdot L}\cdot\frac{\Delta f}{f}$$

6. 噪声在电路中的表示:

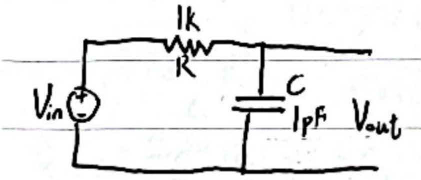
若电路中每个噪声相互独立，则可分别求解噪声到输出端的传输函数，从而等效到输出端，进而求得输入等效噪声 ∠常用输入等效噪声衡量不同系统噪声性能的好坏

计算等效噪声的方法

①将输入电压短接，添加电路中所有噪声源

②计算从噪声源到输出端的传输函数

③将噪声源的功率谱密度与传输函数的平方相乘即可得输出等效噪声的PSD



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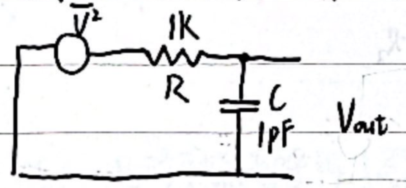


$$\therefore \frac{\overline{V^2_{n,out}}}{df} = 4kT\cdot R\cdot\left|\frac{1}{1+sRC}\right|^2 \qquad SNR = \frac{P_{signal}}{P_{noise}} = \frac{\frac{1}{2}V^2_{out,peak}}{\int_{f_1}^{f_2}\frac{\overline{V^2_{n,out}}}{df}\cdot df}$$

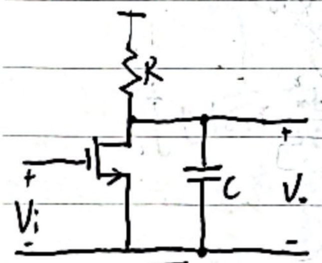
$$\overline{V^2_{n,out,tot}} = \int_0^\infty 4kTR\cdot\left|\frac{1}{1+j2\pi f\cdot RC}\right|^2 df = \frac{kT}{C}$$ 仅与C有关

∴电容值不能过小，否则会影响系统SNR

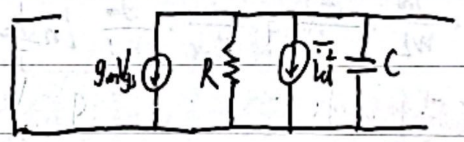
有用的积分公式:

$$\int_0^\infty\left|\frac{1}{1+\frac{s}{\omega_0}}\right|^2 df = \frac{\omega_0}{4} \qquad \int_0^\infty\left|\frac{1}{1+\frac{s}{\omega_0 Q}+\frac{s^2}{\omega_0^2}}\right|^2 df = \frac{\omega_0 Q}{4} \qquad \int_0^\infty\left|\frac{\frac{s}{\omega_0}}{1+\frac{s}{\omega_0 Q}+\frac{s^2}{\omega_0^2}}\right|^2 df = \frac{\omega_0 Q}{4}$$

共源放大器噪声



噪声模型 =>

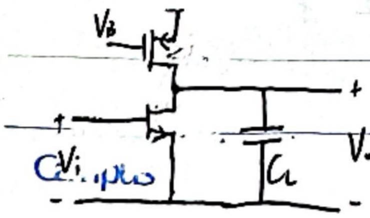


$$V^2 = i^2 Z^2$$

$$\therefore \frac{\overline{V_o^2(f)}}{df} = 4kT\left(\frac{1}{R}+\gamma g_m\right)\cdot\left|\frac{R}{1+j2\pi fRC}\right|^2$$

$$\therefore \overline{V^2_{o,tot}} = \int_0^\infty 4kT\left(\frac{1}{R}+\gamma g_m\right)\cdot\left|\frac{R}{1+j2\pi fRC}\right|^2 df = \frac{kT}{C}(1+\gamma g_m R) = \frac{kT}{C}(1+\gamma|A_v|) = \frac{kT}{C}\cdot\alpha$$

开环MOS管负载的单级放大器



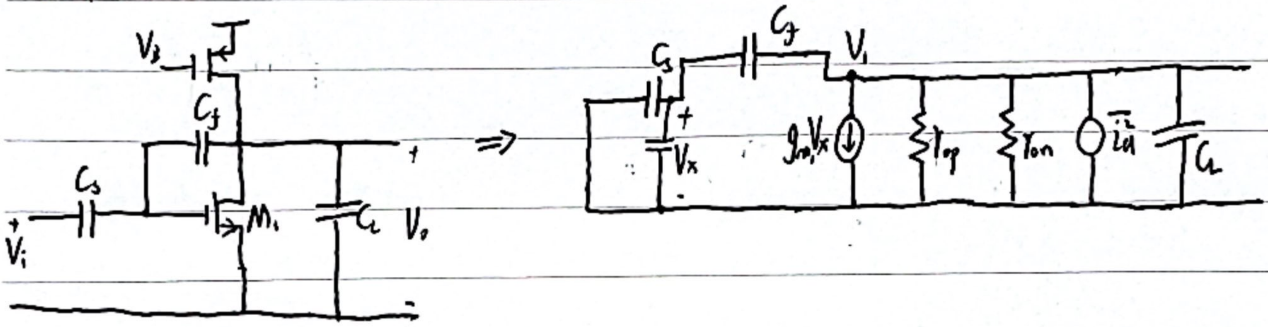
=>

$$\overline{V^2_{o,tot}} = \frac{kT}{C}\cdot\gamma\cdot R_L\cdot(g_{m1}+g_{m2})$$

$$R_L = r_{op} \,//\, r_{on}$$

$\therefore \overline{V^2_{in,tot}} = \frac{\overline{V^2_{o,tot}}}{(g_{m1}R_L)^2} = \frac{KT}{C}\cdot\gamma\cdot\left(1+\frac{g_{m2}}{g_{m1}}\right)\frac{1}{g_{m1}R_L}$ $R_L = r_{op}\|r_{on}$

闭环单级放大器



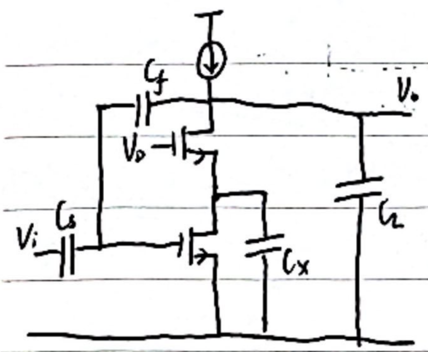
由电压关系可知，$V_x = V_1\cdot\frac{C_f}{C_s+C_x+C_f}$ 记 $\beta=\frac{C_f}{C_s+C_x+C_f}$. $\therefore V_x=\beta V_1$

$\therefore V_1$看入等效阻抗为 $\frac{V_1}{g_m V_x} = \frac{1}{\beta g_m} \ll r_{op}$ or r_{on} $\therefore$ 输出看入阻抗为 $\frac{1}{\beta g_{m1}}$，或者 $R_{L\text{-}close\text{-}loop} = \frac{R_{L\text{-}open\text{-}loop}}{T_s}$

$\therefore \overline{V^2_{o,tot}} = \frac{KT}{C}\cdot\frac{\gamma}{\beta g_{m1}}(g_{m1}+g_{m2}) = \frac{KT}{C}\frac{\gamma}{\beta}\left(1+\frac{g_{m2}}{g_{m1}}\right)$ $C=C_L+C_f(1-\beta)+C_{junction}$ $T_s=\beta g_m R_L$

$\therefore \overline{V^2_{in,tot}} = \frac{KT}{C}\cdot\frac{\gamma}{\beta}\left(1+\frac{g_{m2}}{g_{m1}}\right)\cdot\frac{1}{A^2_{close-loop}}$ $A_{close-loop}=\frac{C_s}{C_f}$

带Cascode单级放大器

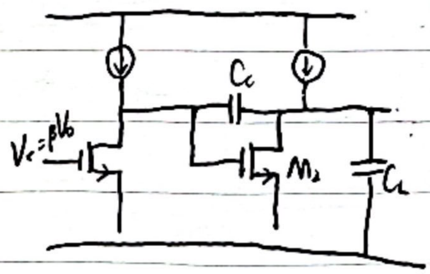


$\overline{V^2_{o,tot}} = \frac{KT}{C}\cdot\frac{1}{\beta}\left(1+\frac{g_{m2}}{g_{m1}}\cdot\frac{W_c}{W_{p2}}\right) = \frac{KT}{C}\frac{1}{\beta}\left(1+\frac{1}{a}\frac{g_{m2}}{g_{m1}}\right)$

$W_c=\frac{\beta g_{m1}}{C_{Ltot}}$ $W_{p2}=\frac{g_{m2}}{C_x}$ $a=\frac{W_{p2}}{W_c}$

$C=C_L+C_f(1-\beta)+C_{junction}$ 减小g_{m2}以减小噪声，但$\frac{g_m}{I_D}$较小意味着V_{ov}较大，减小摆幅

两级放大器



$N_{tot}=\frac{1}{\beta}\frac{KT}{C}\gamma\cdot\frac{1}{1-\frac{\beta g_{m1}}{g_{m2}}}\left[1+\beta\frac{C_c}{C_L}\left(1+\frac{g_{m1}}{g_{m2}}\right)\right]$

为减小输出噪声，需尽可能大的C_c，若C_L较小且β较大，则第二级产生的噪声会很大

总体上所有放大器输出等效噪声都和$\frac{KT}{C}$有关，计算输入等效时又和g_m有关

则 SNR↑6dB → $4C \xrightarrow{BW不变} \frac{1}{4}R \xrightarrow{增益不变} 4g_m \xrightarrow{g_m/I_D不变} 4I_D$ 功耗增加4倍 噪声与功耗tradeoff

- 全差分电路中摆幅变为原来2倍，信号功率变为4倍，噪声2倍，SNR变为原来的2倍，提高3dB，但是付出双倍功耗